PETHIDINE

High-alert drug
Controlled drug

CHEMICAL NAME Ethyl 1-methyl-4-phenylpiperidine-4-carboxylate (as hydrochloride); also called meperidine

CLASS Synthetic opioid agonist (phenylpiperidine)

TRADE NAMES Demerol · Pethidine HCl injection · Dolantin

THERAPEUTIC USES / INDICATIONS

- Moderate to severe acute pain (short-term only).
- Obstetric analgesia during labour.
- Post-operative shivering (small IV doses).
- Pre-operative medication (rarely now).

ADMINISTRATION

- **IM** or **SC** injection; **IV** slowly, diluted, over 3–5 min.
- Oral tablets rarely used (poor bioavailability).
- Controlled-drug rules: two-nurse check and register entry.
- Not recommended for long-term or repeated use.

DOSAGES

- **Adult IM/SC:** 25–100 mg every 4 h; max 400 mg/day.
- **Adult IV:** 25–50 mg slowly, every 4 h.
- **Labour:** 50–100 mg IM, repeated 1–3 h later if needed; max 400 mg/24 h.
- **Children IM:** 0.5–2 mg/kg.
- Reduce dose in elderly and renal impairment.

PHARMACOKINETICS

- **Onset:** IV 5 min; IM 10–15 min. **Peak:** IM 1 h. **Duration:** 2–4 h.
- Metabolised in liver to norpethidine (toxic, long t½, causes seizures).
- Excreted in urine; norpethidine accumulates in renal failure. **t½:** 3–5 h (norpethidine 15–30 h).
- Crosses the placenta.

PHARMACODYNAMICS

- Agonist at **μ opioid receptors** → analgesia, sedation, respiratory depression.
- Has anticholinergic activity → less miosis, tachycardia, dry mouth.
- Shorter action and less constipation than morphine.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting, dizziness, sweating.
- Dry mouth, tachycardia, flushing.
- Sedation, euphoria.

ADVERSE EFFECTS (SERIOUS)

- Respiratory depression — in mother and neonate (if given 1–4 h before birth).
- Seizures, tremor, agitation from norpethidine accumulation.
- Hypotension; serotonin syndrome with MAOIs.
- Dependence and withdrawal.

CONTRAINDICATIONS & CAUTIONS

- Use with MAOIs (or within 14 days); fatal reactions.
- Severe renal impairment; history of seizures.
- Acute respiratory depression; raised intracranial pressure.
- **Caution:** elderly, hypothyroidism, liver disease, sickle-cell disease (repeated doses).

KEY INTERACTIONS

- **MAOIs, linezolid, SSRIs:** serotonin syndrome.
- **Benzodiazepines, alcohol:** additive CNS depression.

NURSING MANAGEMENT

- Check RR, sedation score, SpO_2, BP and pain before and after each dose.
- Withhold if RR <10/min; keep naloxone available.
- Labour: note time of dose; inform neonatal team — baby may need naloxone.
- Watch for tremor, twitching or seizures with repeated doses.
- Follow controlled-drug documentation.

PATIENT EDUCATION

- You may feel drowsy or dizzy; ask for help to walk.
- Report breathing difficulty, shaking or confusion.
- Avoid alcohol and sleeping tablets.
- This drug is for short-term pain only.

KEY POINTS

- Short-acting opioid; toxic metabolite norpethidine.
- Neonatal respiratory depression if given near delivery.
- Avoid in renal failure and with MAOIs.
- Not for chronic pain.

SOURCES CHECKED **[32]** Pethidine 50mg/ml & 100mg/2ml Solution for Injection (UK SmPC) **[33]** Pethidine 50 mg Tablets (UK SmPC) **[34]** Demerol, IM, IV, SC (US label)

Full references p. 432–457